

A Common Lisp Transformer Trainer for a Fixed UMAP Atlas

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Abstract

This document describes a small, dependency-free Common Lisp Transformer that learns to place new medical-evidence observations in an existing two-dimensional UMAP atlas. AWRS–SMC first searches for a useful feature representation and a winning UMAP. The trainer then learns the numerical mapping from the selected features to the coordinates preserved from that winning Common Lisp run. The current implementation trains the entire encoder by backpropagation, serializes the model and preprocessing information as an S-expression, and predicts a new location without fitting UMAP again. Results reported here are a technical pilot, not evidence that the learned atlas has been clinically validated.

1 Purpose and scope

Ordinary UMAP is non-parametric in this project: adding observations requires the embedding to be recomputed if coordinates for the new observations are needed. A parametric mapping instead learns a function

$$f_{\theta} : \mathbb{R}^6 \longrightarrow \mathbb{R}^2,$$

where the input contains the six features selected by AWRS–SMC and the output is a location in the fixed winning atlas. Once f_{θ} has been trained, a new observation can be projected with one forward pass.

This separation is important. AWRS–SMC remains responsible for searching over feature inclusion, transformations, UMAP geometry, clustering quality, and the adjacency preference. The Transformer does not repeat that search. It distills the winning mapping so that it can be reused.

2 From deliberate practice to rapid placement

A useful explanation for a non-specialist is to compare the two stages with deliberate practice. During deliberate practice, a person repeatedly performs a difficult task, receives feedback, and gradually retains a skill that can later be used with less effort. In this project, AWRS–SMC performs the expensive stage. It proposes many alternatives, fits and evaluates candidate UMAPs, and uses the V-measure, the adjacency potential, and the other terms in its score to select a winning atlas. The

resulting annotated examples record both the selected input features and the coordinates produced by that atlas. They therefore provide the feedback from which the Transformer can learn.

Training the Transformer moves part of this computational effort from the time of use to an earlier training phase. Once training is complete, a new study profile is standardized with the stored statistics and passed through the network once. The network predicts its coordinates without rerunning the AWRS–SMC search and without rebuilding the UMAP. In machine-learning terminology, this is a form of *amortized computation*: a substantial cost is paid once so that subsequent predictions are inexpensive. It is also an *out-of-sample parametric mapping*, because one learned function places observations that were not present when the original atlas was made.

The comparison with human learning is an analogy, not a claim that the program learns or understands as a person does. The Transformer compresses a numerical relationship found in its training corpus. It receives no new clinical judgment at prediction time, and rapid placement is trustworthy only to the extent that the training corpus represents the new observation. An input outside the stored feature ranges is an extrapolation and should be flagged for review.

3 Relation to earlier biomedical and clinical work

The broad idea has precedents, although the publications use different models and training objectives. Parametric UMAP replaces the direct optimization of point coordinates with a neural network and explicitly supports fast embedding of new data after training [3]. The closely related *ivis* method learned a parametric map for single-cell expression and mass cytometry data and demonstrated that new observations could be added to an existing embedding [4]. These are biomedical examples, but their observations are cells rather than reports in a clinical meta-analysis.

Transformers have also been trained on heterogeneous clinical and clinicogenomic records before specialization for treatment-response and survival tasks [1]. This supports the general strategy of performing an expensive learning phase once and then reusing the learned representation, but that study did not train a Transformer to reproduce the coordinates of an AWRS–SMC-selected UMAP atlas. A recent preprint studies the specific problem of inserting out-of-sample observations into a fixed UMAP and reports an advantage for parameterized UMAP on increasingly complex data, including medical images [2]. Because it is a 2026 preprint, this evidence should be treated as preliminary.

Thus, fast placement after prior training is an established research direction. The particular teacher–student arrangement used here—AWRS–SMC constructs and scores the atlas, its winning coordinates annotate the corpus, and a small dependency-free Common Lisp Transformer learns that mapping—is a project-specific implementation. It should not be presented as clinical validation or as proof that no earlier system combined comparable components.

4 Annotated training corpus

The corpus builder reads the winning AWRS–SMC result and its preserved Common Lisp coordinates. It reconstructs the winning selected features, freezes the column means and scales, and writes one S-expression record per atlas point. Each record contains an identifier, study group, split, standardized input vector, two-dimensional target, cluster assignment, and medical label.

The pilot contains 90 observations from six study groups. Seventy observations are marked for training and 20 for validation. All observations belonging to one study remain in the same split. The validation study in the present artifact is PALETTE. This prevents windows from one study from appearing on both sides of the evaluation.

The six selected inputs are survival progress, window start, window length, cohort-size context, median progression-free-survival context, and objective response rate, with the exact schema and transformations stored in the corpus. The corpus is numeric rather than textual; “token” below means one named numerical feature.

5 Feature-token Transformer

For feature i with standardized scalar value x_i , the initial token state is

$$h_i^{(0)} = e_i + w_x x_i + b_x,$$

where e_i is a learned feature-identity embedding. Sharing w_x and b_x across positions supplies a common scalar projection, while e_i tells the model which clinical feature the number represents.

The six token states pass through one scaled dot-product self-attention block. For query, key, and value projections,

$$q_i = W_q h_i, \quad k_i = W_k h_i, \quad v_i = W_v h_i,$$

the attention weights and mixed value are

$$a_{ij} = \text{softmax}_j \left(\frac{q_i^\top k_j}{\sqrt{d}} \right), \quad z_i = \sum_j a_{ij} v_j.$$

An output projection, residual connection, and root-mean-square normalization follow attention. A two-layer feed-forward block with a hyperbolic-tangent activation is followed by a second residual connection and normalization. Mean pooling over the six encoded tokens gives a fixed-length representation; a linear regression head returns two coordinates.

The pilot deliberately remains small: one attention block, one attention head, model dimension 8, and feed-forward dimension 12. This controls capacity in a corpus with only 70 training observations.

6 Optimization and reproducibility

The current baseline minimizes coordinate mean squared error,

$$\mathcal{L}_{\text{coordinate}} = \frac{1}{2N} \sum_{n=1}^N \|f_\theta(x_n) - y_n\|_2^2,$$

where y_n is the coordinate pair preserved from the winning Common Lisp UMAP. Scalar automatic differentiation constructs the computation graph during each forward pass. Reverse traversal calculates gradients through scalar projection, self-attention, normalization, the feed-forward block, pooling, and the output head. Adam updates every learned parameter. Thus, this is full

backpropagation—not the frozen-encoder, head-only optimization used by the earlier miniature text Transformer.

Initialization and record ordering are deterministic. The baseline uses seed 20260902, 100 epochs, and learning rate 0.002. The serialized S-expression contains the complete model, feature schema, preprocessing statistics, split policy, and training report. It contains no machine-specific path and requires only SBCL and repository files.

Before full-corpus training, the implementation must overfit one observation. The test verifies that the loss falls by at least four orders of magnitude and below 10^{-6} . It then saves and reloads the weights and verifies that the prediction is unchanged. This is a focused gradient and serialization test; it is not a measure of generalization.

7 Coordinate-only baseline

Table 1 reports the final metrics stored in the trained artifact. Each error is measured in the coordinate system of the winning UMAP.

Table 1: Results after 100 coordinate-only training epochs.

Split	Observations	MSE	RMSE	MAE
Training	70	0.4064	0.6375	0.4939
Validation	20	1.2716	1.1277	1.0213

The validation error is higher than the training error, as expected for an unseen study group. During the fixed 100-epoch run, validation RMSE was approximately 0.958 at epoch 80 and 1.128 at epoch 100. This observation suggests that early stopping may be useful, but the reported baseline retains the prespecified 100 epochs. The validation values were reported and were not used to update model parameters.

8 Running the implementation

From the repository root, the baseline is reproduced with:

```
sbcl --script smc-trainer/train.lisp \
  smc-trainer/corpus/pilot-parametric-umap.sexp \
  smc-trainer/weights/pilot-coordinate-baseline.sexp 100 0.002d0
```

A new observation is projected by supplying its six raw values in the stored feature-schema order:

```
sbcl --script smc-trainer/predict.lisp \
  smc-trainer/weights/pilot-coordinate-baseline.sexp \
  '(0.8 0.1 0.2 2.0 1.0 0.3)'
```

The prediction program reads the means and scales from the trained artifact, standardizes the raw values, and returns an S-expression containing the two coordinates and their coordinate-system identifier.

The corpus and model test suites are run with:

```
sbcl --script vendor/test-cases/run-tests.lisp \  
      smc-trainer/tests.lisp  
sbcl --script vendor/test-cases/run-tests.lisp \  
      smc-trainer/trainer-tests.lisp
```

The current corpus suite passes 1,504 assertions. The model suite verifies an independent product derivative, one-observation overfitting, serialization round-trip, artifact metadata, raw-feature standardization, and finite coordinate prediction.

9 Limitations

The Transformer approximates one fixed atlas and inherits its assumptions and errors. It does not demonstrate that the UMAP geometry is medically correct, and it does not estimate treatment effects. The 90 atlas observations are dependent windows derived from only six study groups. Although the group split prevents direct window leakage, one held-out group is too small a basis for broad claims of generalization.

The preserved target coordinates were constructed using the whole pilot atlas, including the validation study. Consequently, validation measures approximation of a fixed transductive atlas, not independent validation of the atlas’s clinical structure. New observations far outside the feature ranges represented in the corpus are extrapolations and should be marked as such.

10 Next experiment: neighbourhood preservation

Coordinate loss requires the model to reproduce an arbitrary orientation and scale even though the principal scientific interest lies in local relationships. The next experiment will add a neighbourhood term,

$$\mathcal{L} = \mathcal{L}_{\text{coordinate}} + \lambda \mathcal{L}_{\text{neighbourhood}}.$$

Pairs that are neighbours in the winning atlas will be encouraged to remain close in the learned projection; selected non-neighbour pairs will prevent collapse. Several fixed values of λ , including zero, will be compared under the same initialization and split. Coordinate RMSE, nearest-neighbour retention, and cluster agreement will be reported together. The coordinate-only artifact in this document is the required control for that comparison.

References

- [1] Gustavo Arango-Argoty, Elly Kipkogei, Ross Stewart, Gerald J. Sun, Arijit Patra, Ioannis Kagiampakis, and Etai Jacob. Pretrained transformers applied to clinical studies improve

predictions of treatment efficacy and associated biomarkers. *Nature Communications*, 16:2101, 2025. doi: 10.1038/s41467-025-57181-2.

- [2] Mohammad Tariqul Islam and Jason W. Fleischer. On out-of-sample embedding in UMAP. *arXiv preprint arXiv:2606.04451*, 2026. doi: 10.48550/arXiv.2606.04451. Preprint.
- [3] Tim Sainburg, Leland McInnes, and Timothy Q. Gentner. Parametric UMAP embeddings for representation and semisupervised learning. *Neural Computation*, 33(11):2881–2907, 2021. doi: 10.1162/neco_a.01434.
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